

statistics

- the science of collecting, organizing and analyzing data.

- for better decision making.

Types of Statistics

1. Descriptive

- It consists of organizing and summarizing of data.

2. Inferential

- Technique where in we use the data that we have measured to form conclusions.

Sampling Techniques

1. Simple Random Sampling

- randomly pickup elements from population data.
- every element of the population has an equal chance of being selected for sample.

2. Stratified Sampling

- technique where the population is split into non-overlapping groups (strata).

3. Systematic Sampling

- picking every n^{th} individual from the population.

4. Convenience Sampling

- method of choosing individuals who are easiest to reach or contact for data collection.

Variable

- A variable is a property that can take on any value.

Two kinds of variables:

1. Quantitative

2. Qualitative / Categorical

1. Quantitative

- can be measured numerically.

Quantitative (Discrete, Continuous)

Discrete

Continuous.

- specifically have whole numbers (can have decimal (not specified))

- Eg - no. of family members

- Eg : height.

2. Qualitative

- variable that describes a quality or category rather than a numeric value.

- Eg - Gender, Country.

Variable Measurement Scales

- these scales tell us how data can be categorized, ordered or measured.

Types of Scales

1. Nominal Scale : classifies data into distinct categories with no order or ranking.

- Eg : Gender, Blood Type

- Operations allowed : Count, Mode

2. Ordinal Scale : data is placed in order but differences between values are not meaningful.

- Eg - Ranks (1st, 2nd, 3rd).

- Operations allowed - Count, mode, median.

3. Interval Scale : Ordered data with equal intervals, but no true zero.

- Eg : Temperature

- Operations allowed : Add, subtract, mean, median.

4. Ratio Scale : Like interval scale, but with a true zero.

- Eg : Height, Weight.

- All operations allowed.

Measures of Central Tendency

- Refers to the measures used to determine the centre of the distribution of data.

- Mean, median, mode.

Measures of Variation / Dispersion

1. Variance: how spread out the data is.

- variance measures how far each number in a dataset is from the mean, and squares the difference to get a single value.

Why square the difference?

- To avoid negative values cancelling out and to give more weight to larger

differences.

- In short:

more variance = more spread out data.

less variance = data is clustered near the mean.

2. Standard Deviation: how much the data deviates from the mean.

- Standard Deviation tells us how spread out or dispersed the data is from the average (mean), in the same unit as the data.

$$\sigma = \left[\frac{\sum_{i=1}^N (x_i - \mu)^2}{N} \right]^{1/2} \rightarrow \text{For population}$$

For sample, replace N with $(n-1)$.

3. Range :

- it is the difference between minimum and maximum values in a dataset.
- it shows total spread of the data.
- very sensitive to outliers.
- doesn't tell us anything about how values are distributed between min and max.

4. Inter-quartile Range (IQR) :

- it is the range between the first quartile (Q_1) and the third quartile (Q_3).
- it shows the spread of the middle 50% of the data.

$$IQR = Q_3 - Q_1$$

- Not affected by outliers, unlike Range.

- Helps in detecting outliers.

Any value $< Q_1 - 1.5 \times IQR$ or

Any value $> Q_3 + 1.5 \times IQR$

is considered an outlier.

Percentile:

- a percentile is a value below which a certain percentage of observations lie.

Five Number Summary

1. Minimum

2. First Quartile (Q_1)

3. Median

4. Third Quartile (Q_3)

5. Maximum.

Lower Fence $\longleftrightarrow$ Upper Fence

All values of on the left side of Lower Fence and all values on the right side of Upper Fence are outliers.

$$\text{Lower Fence} = Q_1 - 1.5 \times IQR$$

$$\text{Upper Fence} = Q_3 + 1.5 \times IQR$$

Distributions

1. Gaussian / Normal Distribution

- most values are centered around the mean.
- fewer values exist as you move away from the center (mean).
- symmetric bell-shaped curve.
- Mean = Median = Mode.
- spread controlled by Standard Deviation (σ).

$\mu \pm 1\sigma \rightarrow 68\%$ of the data comes in this range.

$\mu \pm 2\sigma \rightarrow 95\%$ of the data comes in this range.

$\mu \pm 3\sigma \rightarrow 99.7\%$ of the data comes in this range.

- many statistical tests assume data is normally distributed.
- it helps in predicting probabilities
- used in z-scores, confidence intervals, hypothesis testing etc.

Standard Normal Distribution (z-Distribution).

- A standard Normal Distribution is a normal distribution that has:

$$\text{mean} = 0$$

$$\text{standard Deviation} = 1$$

z-score

$$z_i = \frac{x_i - \mu}{\sigma}$$

- z-score tells you how many standard deviations a value is above or below the mean.

$z = 0 \rightarrow$ exactly at mean

$z = +1 \rightarrow$ 1 standard deviation above mean.

$z = -1 \rightarrow$ 1 standard deviation below mean.

$z = \pm 2 \rightarrow$ very high/low but still normal.

$z > +3$ or $< -3 \rightarrow$ Extreme values / possible outliers.

- Standard Normal Distribution

- allows comparison across different distributions.

- used to find probabilities using z-tables.

- core of hypothesis testing, confidence intervals etc

Standardization is the process of converting any distribution into a standard Normal Distribution.

mean = 0

standard deviation = 1

- Handles outliers better than normalization.
- Doesn't restrict values to a fixed range.
- Can distort interpretation when data is highly skewed.

Normalization is the process of scaling the data into a fixed range, usually 0 to 1.

- brings everything in 0 to 1 range - models like k-NN, neural networks love it.
- preserves shape of distribution (doesn't distort skewness)
- Sensitive to outliers.
- Works best when only min max are meaningful.

p-value

- it is a number that helps us decide "Should we trust our current belief (the null hypothesis) or is there enough proof to doubt it."
- it is the probability of observing the given data, or something more extreme, assuming that the null hypothesis is true.

$p \leq 0.05 \rightarrow$ reject null hypothesis.

$p > 0.05 \rightarrow$ do not reject

$p \leq 0.01 \rightarrow$ strong rejection

Hypothesis Testing

- it is a way to make decisions or judgements about a population based on a sample of data.
- it helps us know if what we observed is real or just happened by chance.

Basic Terminology

- Null Hypothesis (H_0)

- The Null Hypothesis is like the default belief or status quo.
- You assume the Null Hypothesis is true until you have strong evidence to prove otherwise.
- Null Hypothesis is made in such a way that if it is found to be true, then:
 1. no action is needed.
 2. no decision is needed.
 3. maintain status quo - carry on the work unchanged.
- It is the starting point of Hypothesis Testing. You test your data against H_0 to decide whether to reject it or accept it.

Points to Remember

- H_0 is what you're trying to dispute.
- You never "accept" H_0 - you just fail to reject it if there's not enough evidence.
- Think of H_0 as the "innocent until proven guilty" assumption.

Alternative Hypothesis

- The alternative hypothesis is the statement that suggests a new effect, difference, or relationship exists in the population.
- It's what the researcher is trying to prove or find evidence for.
- This will usually be the opposite of Null

Hypothesis.

- Hypothesis Testing in 6 Steps:

- step-1 : State the Hypotheses (Null and Alternative)

- This is where you define what you're testing.

a. Null Hypothesis

b. Alternate Hypothesis

- step-2 : Choose the Significance Level (α)

- This defines your risk tolerance. - how much chance you're ~~will~~ willing to take of making a mistake (error).

$\alpha = 1 - \text{Confidence Level}$

Types of Errors:

When we do Hypothesis Testing, we either

- Reject the H_0 , or

- Do not reject H_0 .

But sometimes, our decision can be wrong, because we're working with sample data, not the full population.

So 2 types of errors can happen:

1. Type -I Error (False Positive)

- You reject H_0 , even though it's actually true.

- You're saying something is happening, when in reality, nothing is happening.

2. Type-II Error (False Negative)

- You do not reject H_0 , even though its actually False.

- You're saying nothing is happening, when in reality, something is happening.

- Type-I and Type-II errors have a see-saw like or inverse relationship.

★
- Step-3 : Choose the right test and check assumptions.

- You pick a statistical test depending on:

1. Data type : are you testing mean, proportion or variance.

2. Sample size : Small (use t-test) or large (use z-test).

3. One sample or two samples?

4. Standard Deviation known or unknown.

- Step-4 : Calculate the Test Statistic

- Now using formula or commands, we calculate the value of test statistic and the corresponding p-value.

- This step gives us the z-score, t-score etc.

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$\bar{x}$ = sample mean

μ = population mean

σ = population standard deviation.

n = sample size.

- This score tells us how far the sample is from what the null hypothesis expected.

- Step - 5 : Find the p-value and Compare with α .

- Now you check :

- How likely is it to get your test result, if H_0 is True?

- This probability is called the p-value.

- It is the probability of getting results as extreme as the observed ones, assuming that the null hypothesis is True.

If $p\text{-value} \leq \alpha \rightarrow \text{Reject } H_0$

$p\text{-value} > \alpha \rightarrow \text{Do not reject } H_0.$

- Step - 6 : Draw conclusion and Interpret.

- This is where you explain ~~the~~ your result.

- You never say H_0 is 100% true, just that you don't have enough proof to reject it.

- Types of Distributions :

1. Discrete Distributions

- These are used when data comes in countable form.

1. Binomial Distribution

- Used when there are fixed number of trials (n).

- Each trial has only two outcomes $\rightarrow$ Success or Failure.

- Probability of getting a certain number of successes in a fixed number of independent trials.

- Historical probability of the event should be known.

- Poisson Distribution

- used to find the probability of a number of events happening in a fixed time, area, or distance, when the events happen randomly and independently.

- ~~Think~~ Think of it as:

"How many times something will happen in a given interval, if it happens at a known average rate."

- used when the event probability is very small or cannot be calculated.

- Use the historical mean number of occurrences for making the predictions.

2. Continuous Distributions

- Used when data is in continuous form.

i.e., the possibility of a decimal point cannot be dismissed.

1. Exponential Distribution

- used when events happen randomly and independently, at a constant average rate.

- used where you want to model the time interval between two successive events.

- it tells us:

"How long you'll have to wait before the next event happens."

- avg time between the events is known.

2. Student's T Distribution

- used when sample size is less than 30.

- used when you're working with small sample sizes and you don't know the population standard deviation.

- as the sample size increases, the probability values from the T-Distribution will get closer and closer to the values from a normal distribution.

- T-Distribution can be used in all circumstances irrespective of the sample size.

- Central Limit Theorem (CLT)

- it says:

"If you take many random samples from a population and calculate their means, then the distribution of those means will look like a normal distribution, no matter what the shape of the original population is, as long as the sample size is big enough (usually ≥ 30).

- in simple words:

- original data can be weird, skewed, non-normal.
- take multiple random samples
- calculate mean of each sample.
- Those means will form a bell-shaped curve.

- Why is CLT important?

- You can apply normal distribution formulas even if your data is not normal.
- Helps in Hypothesis Testing, Confidence Intervals etc.
- Allows you to work with small samples from a big population.

- Standard Deviation of the sample means would be equal to

$$\text{Std (sample means)} = \frac{\text{Std (population)}}{(\text{sample size})^{**} 0.5}$$

- Bias

- Bias is a consistent, unfair influence that pushes results in a particular direction.

- Kaa bar result galat aa raha hai, ek hi direction mein, because something is wrong in the process.

- Eg: - A measuring scale that's always 2 cm too short $\rightarrow$ every height you record will be less than equal.

- A teacher giving more marks to students they like $\rightarrow$ biased grading.

- Types of Bias

1. Selection Bias : Choosing wrong or non-random sample.

2. Response Bias : People lie or answer incorrectly.

3. Measurement Bias : Wrong instrument/method used.

- Chance Variation

- Chance Variation is the random ups and downs that happen naturally in data.

- meaning:

"kabhi zyada, kabhi kam - but no fixed pattern. This is just luck, randomness, or sampling error."

- Chance Variation can be completely removed only if the data is collected from the entire population i.e.

sample size = population size.

- Final Line

- Bias is like a crooked gun - always shoots off-target..

- Chance Variation is like wind blowing the bullet randomly - sometimes left, sometimes right.

- Confidence Level

- It tells you how sure you are that your sample result reflects the true population.

- Eg : You take a survey of 100 students, and find average height = 165 cm.

You say : "I'm 95% ~~sure~~ confident that

the true average height of all students is between 163 cm and 167 cm.

- This 95% is your confidence level.

- The 163 - 167 cm range is your confidence interval.

- Confidence Level = 100% only when

sample size = population size.

- Confidence Interval

- It is the range in which we expect the true population value (like mean or proportion) to lie, based on sample data.

- Size of Interval depends on:

1. Confidence Level : As the CL increases, the size of the interval increases and vice versa.

- Sample Size : As the sample size increases, size of the interval decreases and vice versa.

- Variation in the data (std. Deviation).

- As the variation decreases, the size of the interval decreases and vice versa.

- Left Tail Test

- It is a type of one-tail test where you're checking if a value is significantly less than something.

- Error is on the left side of the value.

H_0 : The parameter is greater than or equal to some claim value.

H_1 : The parameter is less than the value.

- Use it when your suspicion or claim is that something has decreased (mean, proportion etc.).

Eg - Has the average salary gone down?

- Is the new machine producing fewer items than before?

- Has the average marks of students dropped?

- Right Tail Test

- used when you want to check if a value is significantly greater than some claim value.
- Error is on the right side of the value.

H_0 : Parameter is less than or equal to a value.

H_1 : Parameter is greater than the value.

- Use it when you suspect the average proportion or rate has increased.

Eg - Has the average speed increased?

- Is the new fertilizer giving more crop yield?
- Has the success rate improved after a new strategy?

- Two -Tail Test

- used when you're checking if a value is significantly different - either greater than or less than some claim value.
- You don't care which direction the change is, you just want to know if there is any change at all.

H_0 : The parameter is equal to a claim value.

H_1 : The parameter is not equal to the claim value.

- Use it when you're checking for any change (increase or decrease).

Eg - Is the average temperature changing?

- Is the exam pass rate different this year?

- Tests for Continuous Data (Numerical Data)

- 1 sample z-test / 1 sample t-test

- used when we wish to compare the mean of the sample with a claim value and establish whether the difference between them is statistically significant or not.

z-test

- used when standard Deviation of population is known.

- sample size > 30 .

- more accurate when population info is complete.

t-test

- Population standard Deviation unknown.

- sample size ≤ 30 .

- more accurate when only sample data is available.

- 2 sample t-test

- used when difference of averages of two independent samples has to be compared against a claim value and establish whether the difference is statistically significant or not.

- average of the two samples is taken first, then their difference is compared against a claim value.

- Eg :

$$H_0 : \mu_1 - \mu_2 \geq \text{claim value}$$

$$H_1 : \mu_1 - \mu_2 < \text{claim value}$$

- Paired t-test

- before vs after data for the same group of people or items, and you want to check if the avg change in before and after values of an event is statistically significant or not.

- sample size needs to be the same and in the same order.

- first difference is taken of the paired values and then average is taken of the paired values and compared with a claim value.

- What is ANOVA?

- ANOVA stands for Analysis of Variance.

- It is a statistical method used to compare the means of two or more groups to see if there is a significant difference between them.

- If you want to compare:

- 2 groups $\rightarrow$ you can use a t-test.

- But if you have 3 or more groups, running multiple t-tests increases error.

Eg- Let's say you have 3 groups A, B, C.

You decide to compare them using t-tests:

A vs B

B vs C

A vs C

That's 3 t-tests in total.

Each t-test has a 5% chance of a false positive α (Type-I error) if the null hypothesis is true.

- The probability of at least one false positive is

$$\text{error rate} = 1 - (1 - \alpha)^k$$

$$\alpha = 0.05 \text{ or } 5\%$$

$$k = 3 \text{ (no. of comparisons)}$$

Error rate $\approx 14.3\%$

So, instead of a 5% chance of error, you now have 14.3% chance of making a wrong conclusion.

- And this gets worse with more groups.

For 5 groups $\rightarrow 10$ comparisons $\rightarrow 1 - (1 - 0.05)^{10} \approx 40\%$ error chance.

- ANOVA tests all group means together in one test, keeping the error rate at 5%.

- One Way ANOVA

- used when we have more than 2 samples.

This test will compare the means of all pairs of samples.

H_0 : means of all pairs of samples are equal.

i.e. $\text{mean}(A, B) = \text{mean}(B, C) = \text{mean}(A, C)$

H_1 : means of at least one pair of samples are not equal.

- Two Way ANOVA or Multiple ANOVA.

used when there are two or more independent factors influencing the response or one dependent factor, and to check if there's any interaction between the two factors.

- Use two-way ANOVA when:

1. You have one numerical outcome (dependent variable).
2. You have two categorical variables (factors) each with 2 or more groups.

3. You want to see:

- If factor A affects the outcome.
- If factor B affects the outcome.
- If A and B together (interaction) affect the outcome.

- One Variance Test

- it is used to check if the variance (or standard deviation) of a population is equal to a specific value.

- Used when:

1. you have numerical (continuous) data.
2. you want to check if the spread or

variability in the data is equal to a given value.

- Two Variance Test
 - used to compare the variances of two different samples to see if they are equal or significantly different.

Used when:

1. You have two independent samples.
2. Both have continuous data.
3. You want to test if their variances (or standard deviation) are the same.

For Discrete Data

One Proportion Test

- used when the proportion calculated from a sample has to be compared against a claim value.

H_0 : proportion = claim value.

H_1 : proportion $\neq$ claim value

Eg - A company claims 70% of users are satisfied.
You survey 100 users and find 62 are satisfied.

Want to test: Is the satisfaction rate really

- Two Proportion Test

- used when the difference of proportions from two independent samples has to be compared against a claim value.

Eg: Out of 100 men, 60 prefer product A.

Out of 120 women, 66 prefer product A.

Test: Is there a difference in proportions?

$$H_0: P_1 - P_2 = 0$$

$$P_1 = P_2$$

$$H_1: P_1 \neq P_2$$

- Chi-Square Test for Goodness of Fit

- used to determine whether a variable is likely to come from a specified distribution or not. It is ~~not~~ often used to evaluate ~~when~~ whether sample data is representative of the full population.
- to establish whether counts obtained from a sample follow the pattern which has been claimed for the different categories or not.

H_0 : counts follow the pattern

H_1 : there is at least one category where the counts do not follow the pattern